

The Entity relationship model is a conceptual model for designing a database. This model represents the logical structure of a database, including entities, their attributes, and relationships between them.

Entity:- An object that is stored as data  
eg: Student, course or company

Attribute: Properties that describe an entity.  
eg: student\_id, courseName or employee\_email

Relationship: A connection between entities  
eg: student enrolls in a course

ER model in database design process:-

- 1) Gather the requirements by asking questions to the database users
- 2) Create a logical or conceptual design of the database. this is where ER model plays a role. It is the most used graphical representation of the conceptual design of a database

Entity:-

An entity represents a real world object, concept or thing about which data is stored in a database. It act as a building block of a database. Tables in database represents these entities.

Example of entities:-

Real world objects : person, car, employees etc.

Concepts: course, Event, Reservation etc.

Things: products, Document, device etc.

Types of entity:-

1) Strong entity:-

A strong entity is a type of entity that has a key attribute that can uniquely identify each instance of the entity. A strong entity does not depend on any other entity in the schema for its identification. It has a primary key that ensures its uniqueness and is represented by a rectangle in ER diagram

2) Weak entity:-

A weak entity cannot be uniquely identified by its own attributes alone. It depends on a strong entity to be identified. A weak entity is associated with an identifying entity(strong entity), which helps in its identification. A weak is represented by double rectangle.

The relationship between the weak entity type and its indentifying

strong entity type is called identifying relationship(weak relationship) and it is represented by a double diamond.

### 3) Attributes in ER model:-

Attributes are the properties that define the entity type.

For example:-

Roll\_no, name, dob, age, address and mobile\_no are the attributes that define entity type student.

Attributes are represented by oval.

### Types of attributes:-

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#### 1) Key attribute:-

Attribute which uniquely identifies each entity in the entity set is called the key attribute.

For example :- roll\_number, student\_id, college\_id, employee\_id etc

The key attribute is represented by an oval with an underline.

#### 2) Composite attribute:-

An attribute composed of many other attributes is called a composite attribute.

for example, the address attribute of the student or customer entity type consists of street, city, state and country.

In ER diagram the composite attribute is represented by an oval comprising of ovals.

#### 3) Multivalued attributes:-

An attribute consisting of more than one value for a given entity.

for example phone\_no(can be more than one for a given student).

In ER diagram, a multivalued attribute is represented by a double oval

#### 4) Derived attribute:-

Derived attribute an attribute that can be derived from other attributes of the entity type is known as a derived attribute.

eg:- age (can be derived from DOB).

In ER diagram, the derived attribute is represented by a dashed oval

### Relationship type and relationship set:-

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A relationship type represents the associations between entity

types.

For example "Enrolled in " is a relationship type that exists between entity type Student and Course. In ER diagram, the relationship type is represented by a diamond and connecting the entities with lines.

A set of relationships of the same type is known as a relationship set. The above relationship set depicts S1 as enrolled in C2, S2 enrolled in C1 and S3 enrolled in C3.

Degree of relationship set:-

The number of different entity sets participating in a relationship set is called the degree of relationship set.

1) Unary/ Recursive Relationship:-

When there is only ONE entity set participating in a relation, the relationship is called unary relationship.

for eg:- one person is married to only one person

2) Binary relationship:-

When there are Two entities set participating in a relationship, the relationship is called a binary relationship.

For example a student is enrolled in a course

3) Ternary Relationship:-

When there are three entity sets participating in a relationship, the relationship is called ternary relationship.

4) N-ary relationship:-

When there are N entities set participating in a relationship the relationship is called an n-ary relationship

Cardinality in ER model:-

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The maximum number of times an entity set participates in a relationship is known as cardinality.

Types of Cardinality:-

1) One to One:-

When each entity set can take part only once in the relationship the cardinality is one-to-one.

2) One-to-many:-

In a one to many relationship, one entity can be associated with multiple entities.

For example:-

3) Many to one:- opposite of (one to many)

4) Many to Many:-

Many to Many is when entities in all entity sets can take part more than once in the relationship cardinality is many to many

How to draw an ER diagram:-

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1) Identify entities:-

The very first step is to identify all the entities.

Represent those entities in a Rectangle and label them accordingly.

2) Identify relationships:-

the next step is to identify relationship between them and represent them accordingly using the Diamond shape. Ensure that relationships are not directly connected to each other.

3) Add attributes:-

Attach attributes to the entities by using ovals.

Each entity can have multiple attributes(such as name, age etc) which are connected to the respective entity.

4) Define primary keys:-

Assign primary key to each entity.

5) Remove redundancies:-

review the diagram and eliminate unnecessary or repetitive entities and relationships

6) Review for clarity.